



March 25, 2021

Mr. Doug Langworthy
City of Cadillac WWTP
200 N. Lake Street
Cadillac, Michigan 49601

Reference: 0550183.0106

Subject: Whole Effluent Toxicity Test Results

Dear Doug,

Enclosed please find the final results of the following Chronic Toxicity Tests performed on samples of the City of Cadillac WWTP Outfall 001A effluent.

- 9 March 2021, Chronic *Ceriodaphnia dubia* Toxicity Test
- 9 March 2021, Chronic *Pimephales promelas* Toxicity Test

If you have any questions concerning this report or if I can be of any further assistance to you, please feel free to contact me at (616) 738-7308 or via e-mail at bruce.rabe@erm.com.

Yours sincerely,

Bruce A. Rabe
Director, Aquatic Toxicology Laboratory

BAR:ks

Enclosures: Whole Effluent Toxicity Test Report

cc: Cindy Tomaszewski
File

FINAL REPORT

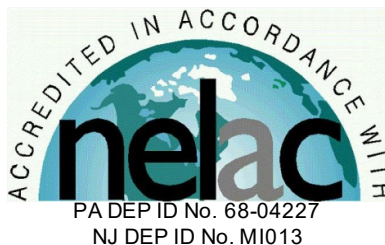
Chronic Toxicity Test
Freshwater Invertebrate,
Ceriodaphnia dubia
EPA Test Method 1002.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 9 – 15 March 2021

Laboratory I.D. Number: 030921-6



Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424



Test Overview



Permittee: City of Cadillac WWTP
Location: 1121 Plett Rd.
Cadillac, MI 49601
Contact: Mr. Doug Langworthy
Telephone #: 231.878.4437

NPDES Permit #: MI0020257
Permit Requirements: Reporting Criteria – 1.0 TUa, 1.0 TUc
Test Sample: Outfall 001A
Receiving Water: Clam River

Testing Date: 9 – 15 March 2021

Sample Date(s): 8 March 2021
10 March 2021
12 March 2021

Test/Method: Daphnid, *Ceriodaphnia dubia*,
Survival and Reproduction Test
EPA 821-R-02-013 Method
1002.0.

QC Objectives: Test data met all test
acceptability criteria, except
where noted below.

Data Qualifiers: None

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Reproduction (Average Young/Female)
Control	100	30.2
13	100	38.5
25	100	32.8
50	100	36.6
76	100	33.6
100	100	27.7

TEST RESULTS

48-Hour LC ₅₀	>100%
NOEC	100%
LOEC	>100%
ChV	>100%
MSDp	29.2%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	0

TEST CONCLUSION

In accordance with the NPDES permit requirement for the City of Cadillac WWTP, this toxicity test did not exhibit either acute or chronic toxicity.

Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0550183.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777



ERM Testing Method

Ceriodaphnia dubia – Survival and Reproduction Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Ceriodaphnia dubia used to initiate this test were obtained from individual, in-house cultures and were less than 24-hours old, and had an age range of 0 to 8 hours at test initiation. Test organisms used to initiate this test were released from adults which met acceptable performance criteria (i.e., ≥15 young/surviving female within 3 broods and obtained from a brood of at least 8 young) and were maintained in reconstituted moderately hard water prior to test initiation.

The *Ceriodaphnia dubia* test was conducted using 30-milliliter (mL) disposable polystyrene containers containing 15 mL of control water or test solution. One *Ceriodaphnia dubia* was added to each test chamber with ten replicate chambers per treatment. Each *Ceriodaphnia dubia* test chamber was fed a 0.2-mL suspension consisting of yeast-Cerophyll-trout chow (YCT) and green algae (*Raphidocelis subcapitata*) mixture daily.

The test solutions were renewed daily during the exposure by transferring the adult daphnid, by way of a wide bore pipette, into fresh control water or test solution.

Percent survival of exposed *Ceriodaphnia dubia* was determined by inspecting for adult mortality daily. Mortality was defined as no body or appendage movement after gentle prodding. Production of young was also determined by daily inspections and enumeration. When 60 percent of the surviving females in the control treatment produced three broods, mean reproduction was determined by calculating the average number of live young produced per female for each treatment.

The test was conducted at a temperature of $25 \pm 1^{\circ}\text{C}$ under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Ceriodaphnia dubia* survival and reproduction. An NOEC is defined as the highest effluent concentration that does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration that does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration. Significant differences in *Ceriodaphnia dubia* survival were determined using the Fisher's Exact Test.

Prior to the determination of any significant differences in *Ceriodaphnia dubia* reproduction, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or reproduction) and converted to a chronic toxic unit (TUc) for reporting purposes by 100/ChV, unless the LOEC was greater than 100 percent in which case the TUc was reported as 0.

To evaluate acute toxicity, a 48-hour LC₅₀ and corresponding 95 percent confidence interval was also calculated, where possible. The LC₅₀ value estimate was determined by using one of the following statistical methods: graphical, Spearman-Kärber, Trimmed Spearman-Kärber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 48-hour LC₅₀ value was converted to an acute toxic unit (TUa) by 100/LC₅₀.

For 48-hour LC₅₀ values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 percent to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 percent to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.7.6 software program.

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms and the precision of the testing procedure. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC₂₅) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC₂₅ of a particular reference toxicant test does not fall within the expected range of $\pm$ two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

**Environmental
Resources
Management**

***Ceriodaphnia dubia* - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation**

Table 1
Page 1 of 1

Permittee/Client: City of Cadillac WWTP

Effluent/Location: Outfall 001A

Lab I.D.#: 030921-6

Beginning Date: 03/04/21

Ending Date: 03/15/21

Time: 1400

Time: 1630

Control/Dilution Water: RMHW

Organism Batch #: 43-21

Organism Age: 16-25 hrs
24 hrs 03/17/21

QC Review: 5/12

QC Review Date: 03/19/21

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	030921-6	031121-3	031321-8	26-21	33-21	--
Temperature	° C	1	1	3	--	--	--
Dissolved Oxygen	mg / L	10.6	11.2	10.7	--	--	--
pH	S.U.	7.1	7.5	7.4	8.2	8.1	--
Conductivity	umhos/cm	1056	942	959	316	317	--
Alkalinity	mg / L CaCO ₃	66	94	120	62	62	--
Hardness	mg / L CaCO ₃	240	220	240	86	84	--
Total Ammonia	mg / L NH ₃	0.06	0.07	0.09	--	--	--
Total Residual Chlorine	mg / L Cl ₂	20.01	20.01	20.01	20.01	20.01	--
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	VS	MT	SPR	TMB	SPR	--

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)	Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200	0	TMB	1	26-21
13%	156	1044	1	TMB	1	26-21
25%	300	900	2	MT	2	26-21
50%	600	600	3	TMB	2	26-21
76%	912	288	4	DM	3	26-21
100%	1200	0	5	BAR	3	33-21
			6	MT	3	33-21
			7			

**Environmental
Resources
Management**

**Ceriodaphnia dubia - Chronic Toxicity Test
Water Quality Data**

Table 2
Page 1 of 1

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 030921-6

Water Quality Data:

Dissolved Oxygen (mg/L)														
Day														
Meter #	3	3	3	3	3	5	3	3	3	3	3	3	3	
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.5	7.9	8.4	8.0	8.1	8.4	8.2	8.2	8.5	8.1	8.4	8.5	8.4	
13%	8.5	8.0	8.4	8.1	8.2	8.4	8.2	8.2	8.4	8.2	8.5	8.5	8.5	
25%	8.5	8.1	8.4	8.1	8.1	8.4	8.3	8.3	8.4	8.1	8.5	8.5	8.5	
50%	8.4	8.1	8.4	8.1	8.1	8.4	8.2	8.1	8.4	8.1	8.5	8.5	8.4	
76%	8.4	8.1	8.4	8.1	8.2	8.4	8.2	8.2	8.4	8.1	8.5	8.5	8.4	
100%	8.4	8.0	8.4	8.1	8.3	8.2	8.2	8.0	8.4	8.0	8.5	8.4	8.4	
pH (S.U.)														
Day														
Meter #	9	9	9	10	10	9	10	10	10	10	10	9	9	
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.1	8.1	8.0	8.2	8.1	8.1	8.0	8.0	8.1	8.0	8.0	8.2	8.0	
13%	--	8.3	--	8.1	--	8.2	--	8.1	--	8.0	--	8.3	--	
25%	--	8.3	--	8.1	--	8.3	--	8.1	--	8.1	--	8.3	--	
50%	--	8.3	--	8.0	--	8.4	--	8.1	--	8.2	--	8.4	--	
76%	--	8.2	--	8.0	--	8.3	--	8.1	--	8.2	--	8.4	--	
100%	7.4	8.2	7.3	7.9	7.7	8.3	7.7	8.2	7.6	8.2	7.4	8.4	7.5	
Conductivity (umhos / cm)														
Day														
Meter #	4	--	4	--	4	--	4	--	4	--	4	--	4	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	311	--	313	--	312	--	315	--	317	--	321	--	317	--
13%	387	--	388	--	387	--	390	--	403	--	409	--	401	--
25%	455	--	455	--	456	--	455	--	481	--	488	--	475	--
50%	595	--	595	--	595	--	596	--	640	--	651	--	640	--
76%	763	--	764	--	764	--	741	--	832	--	820	--	804	--
100%	894	--	876	--	844	--	870	--	984	--	1000	--	986	--
Temperature (°C)														
Day														
Meter #	3	3	3	3	3	5	3	3	3	3	3	3	3	
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	24	24	24	24	24	24	24	25	24	25	24	24	
13%	24	24	24	24	24	24	24	24	25	24	25	24	24	
25%	25	24	24	24	25	24	25	24	25	24	25	24	24	
50%	25	24	24	24	25	24	25	24	25	24	25	24	24	
76%	26	24	24	24	25	24	25	24	25	24	25	24	24	
100%	26	24	24	24	25	24	26	24	25	24	25	24	24	
I = Initial Chemistry F = Final Chemistry														

Note: D.O. meter also used for temperature measurement unless otherwise noted.

Environmental
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Ceriodaphnia dubia - Chronic Toxicity Test
Survival and Reproduction Data

Table 3
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Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 030921-6

Treatment (% Effluent)	Day No.	Replicate										Average Young/ Female	Number of Live Adults (% Sur.)	Average Young/ Female % CV
		1	2	3	4	5	6	7	8	9	10			
Control	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	6	5	--	7	--	--	--	4	--	--		10	
	4	--	--	6	--	5	8	5	--	7	6		10	
	5	11	8	18	14	12	13	10	9	11	11		10	
	6	--	--	15	20	20	19	14	16	18	14		10	
	7	S/R 03/17/21											10	
Totals:		10	13	31	41	37	38	29	29	36	31	30.2	(100)	29.9
# Broods (% 3rd Brood)		2	2	3	3	3	3	3	3	3	3	(80)		
13%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	5	--	4	5	--	--	--	--	--		10	
	4	5	--	7	--	--	5	5	4	6	7		10	
	5	10	13	12	12	14	13	13	10	10	14		10	
	6	14	21	24	24	21	21	20	22	19	23		10	
	7	S/R 03/17/21											10	
Totals:		19	39	43	42	40	39	38	36	35	44	37.5	(100)	11.4
25%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	5	--	--	--	--	6		10	
	4	6	5	6	4	--	5	6	1	6	--		10	
	5	12	14	11	16	16	10	10	12	16	12		10	
	6	19	24	21	21	23	14	13	20	--	22		10	
	7	S/R 03/17/21											10	
Totals:		18	43	33	37	44	29	29	33	22	40	32.8	(100)	26.2
50%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	6	5	--	--	5	--	--		10	
	4	5	7	6	--	--	6	5	--	6	6		10	
	5	12	16	15	13	13	13	11	10	15	13		10	
	6	19	21	21	18	19	24	18	19	20	--		10	
	7	S/R 03/17/21											10	
Totals:		36	44	42	37	36	43	34	34	41	19	36.6	(100)	19.7
76%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	6	4	--	--	--	--	2		10	
	4	8	6	6	--	--	5	6	5	6	--		10	
	5	10	11	12	11	16	12	12	10	11	11		10	
	6	18	--	23	22	22	14	24	18	13	12		10	
	7	S/R 03/17/21											10	
Totals:		36	17	41	39	42	31	42	33	30	25	33.6	(100)	24.3
100%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	4	--	--	--	--	--	--		10	
	4	5	--	7	--	6	6	5	5	4	5		10	
	5	11	--	12	10	14	12	10	8	9	--		10	
	6	5	--	16	20	18	21	21	16	14	11		10	
	7	S/R 03/17/21											10	
Totals:		21	0	37	34	38	39	36	29	27	16	27.7	(100)	44.7

X = DEAD ADULT 1X = DEAD ADULT, ONE YOUNG PRODUCED BEFORE DEATH -- = NO YOUNG RECORDED
(E) = ABORTED EMBRYOS /EGGS (1) = ONE DEAD YOUNG (S) = SPLIT BROOD * = 4th BROOD EXCLUDED FROM TOTAL

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 030921-6

Brood Board Information:

Brood Board Information:											
Replicate	1	2	3	4	5	6	7	8	9	10	Brood Board Date: 2/24
Chamber Number	14	24	25	29	32	40	45	59	54	4	Young Age Range: 16-21.5 hours

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
YCT Batch #:	04-21	04-21	04-21	04-21	04-21	05-21	05-21	
Algae Batch #:	05-21	05-21	05-21	05-21	05-21	05-21	05-21	
Observation Time:	1400	1330	1430	1330	1300	1130	1630	
Initials:	mn	mt	ms	mn	DAH	DAH	mt	
Date:	03/04/21	03/10/21	03/11/21	03/12/21	03/13/21	03/14/21	03/15/21	

Comment Section:

[illegible]

CETIS Analytical Report

Report Date: 17 Mar-21 14:16 (p 1 of 2)
Test Code/ID: 5A42F9F0 / 15-1433-8800

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 21-0060-1414	Endpoint: Reproduction	CETIS Version: CETISv1.9.7
Analyzed: 17 Mar-21 14:15	Analysis: Nonparametric-Control vs Treatments	Status Level: 1
Edlt Date: 17 Mar-21 14:13	MD5 Hash: AAC862F22888D5CE9B9B4119565E83E	Editor ID: 009-972-616-9
Batch ID: 07-4990-5985	Test Type: Reproduction-Survival (7d)	Analyst: Lab Tech
Start Date: 09 Mar-21 14:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 15 Mar-21 16:30	Species: Ceriodaphnia dubia	Brine:
Test Length: 6d 2h	Taxon: Branchiopoda	Source: In-House Culture Age: <24
Sample ID: 08-5283-5143	Code: 32D53B47	Project: Wet Compliance Test
Sample Date: 08 Mar-21 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 09 Mar-21 11:00	CAS (PC):	Station: Outfall 001A
Sample Age: 31h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	MSDu	PMSD
Untransformed	C > T	100	>100	---	10	8.832	29.24%

Steel Many-One Rank Sum Test

Per MDEGLE

Control	vs	Conc-%	Test Stat	Critical	Ties	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	136	75	3	18	CDF	0.9999	Non-Significant Effect
		25	113.5	75	2	18	CDF	0.9590	Non-Significant Effect
		50	129	75	3	18	CDF	0.9992	Non-Significant Effect
		76	118.5	75	4	18	CDF	0.9860	Non-Significant Effect
		100	100.5	75	4	18	CDF	0.7129	Non-Significant Effect

Test Acceptability Criteria

TAC Limits

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	30.2	15	>>	Yes	Passes Criteria
PMSD	0.2924	0.13	0.47	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	792.133	158.427	5	2.129	0.0758	Non-Significant Effect
Error	4018.6	74.4185	54			
Total	4810.73		59			

ANOVA Assumptions Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variance	Bartlett Equality of Variance Test	8.713	15.09	0.1211	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.9192	0.9459	0.0007	Non-Normal Distribution

Reproduction Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	10	30.2	23.75	36.65	31	13	41	2.851	29.85%	0.00%
13		10	38.5	35.35	41.65	39	29	44	1.392	11.44%	-27.48%
25		10	32.8	26.66	38.94	33	18	44	2.715	26.18%	-8.61%
50		10	36.6	31.44	41.76	36.5	19	44	2.281	19.71%	-21.19%
76		10	33.6	27.76	39.44	34.5	17	42	2.583	24.31%	-11.26%
100		10	27.7	18.84	36.56	31.5	0	39	3.916	44.70%	8.28%

Reproduction Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	17	13	31	41	37	38	29	29	36	31
13		29	39	43	42	40	39	38	36	35	44
25		18	43	33	37	44	29	29	33	22	40
50		36	44	42	37	36	43	34	34	41	19
76		36	17	41	39	42	31	42	33	30	25
100		21	0	37	34	38	39	36	29	27	16

CETIS Analytical Report

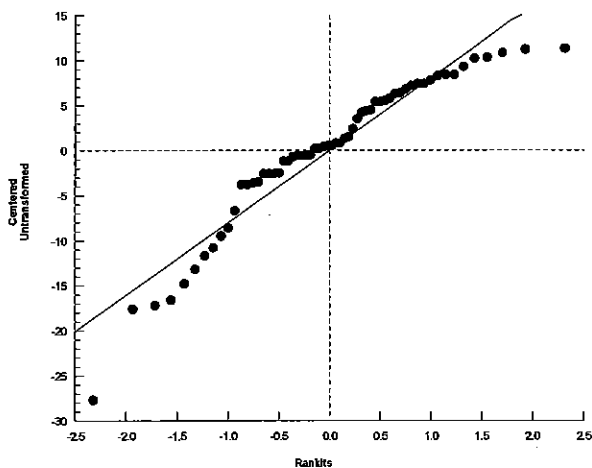
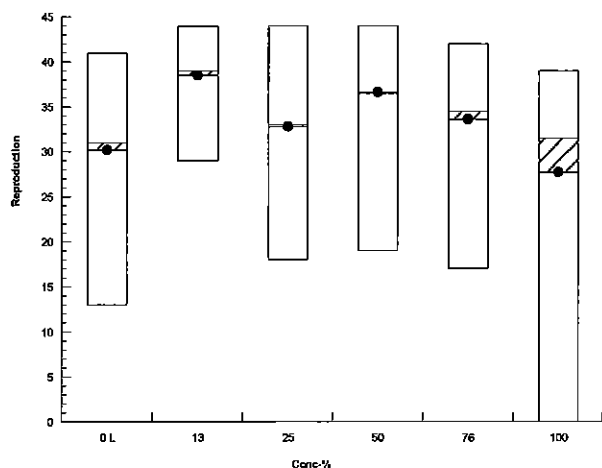
Report Date: 17 Mar-21 14:16 (p 2 of 2)
Test Code/ID: 5A42F9F0 / 15-1433-8800

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 21-0060-1414	Endpoint: Reproduction	CETIS Version: CETISv1.9.7
Analyzed: 17 Mar-21 14:15	Analysis: Nonparametric-Control vs Treatments	Status Level: 1
Edit Date: 17 Mar-21 14:13	MD5 Hash: AAC862F228888D5CE9B4119565E83E	Editor ID: 009-972-616-9

Graphics





3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

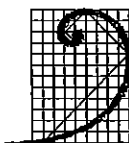
CLIENT NAME:	SAMPLER	PHONE NUMBER:						
City of Cadillac 1121 Platt Rd Cadillac, MI	TZ	231-775-2841						
SAMPLE DESCRIPTION (i.e. Outfall 001)	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)					
Tert.	pH=6.89 s.u. NH ₃ =.04 mg/L	030921-6	Temp. (°C) ☒ On Ice	D.O. 10.6 mg/L	pH 7.1 s.u.	Cond/Sal 1056	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
	pH= NH ₃ =		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt	
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product	Test Type: ☒ Acute ☒ Chronic ☐ Other	Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☒ Fathead minnow (<i>Pimephales promelas</i>) Rainbow Trout (<i>Oncorhynchus mykiss</i>) Sheepshead minnow (<i>Cyprinodon variegatus</i>) Silverside minnow (<i>Menidia beryllina</i>) Other (write in comments section)					Americamysis bahia Hyalella azteca Chironomus dilutus
COMMENT SECTION:								

SAMPLE TRANSFERS

SAMPLE TRANSFERS					
RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
Tarsha Layton / City of Cadillac	3-8-21	2 PM	Mary Wisniewski / ERM	3/9/21	1100

* See Instructions for Sample Collection on Back of Sheet

December 2020



ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

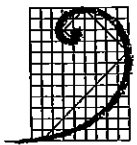
CLIENT NAME:		City of Cadillac		SAMPLER		TL		231-775-2841		
ADDRESS:		1121 Plett Rd Cadillac, MI		PHONE NUMBER:						
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
	3-9-21 3-10-21	7AM 7AM	Comp	2gal	pH= 7.23 s.u. NH ₃ = .055 mg/L	031121-3	Temp. (°C) On Ice	D.O. mg/L 11.2	pH s.u. 7.5	Cond/Sal 942
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice	D.O. mg/L	pH s.u.	Cond/Sal
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice	D.O. mg/L	pH s.u.	Cond/Sal
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice	D.O. mg/L	pH s.u.	Cond/Sal
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice	D.O. mg/L	pH s.u.	Cond/Sal
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice	D.O. mg/L	pH s.u.	Cond/Sal
ANALYSES REQUESTED [check item(s)]	Test Material:		Test Type:		Test Species:					
	☒ Water/Wastewater ☐ Sediment ☐ Product		☒ Acute ☒ Chronic ☐ Other		☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☒ Fathead minnow (Pimephales promelas)		☐ Rainbow Trout (Oncorhynchus mykiss) ☐ Sheepshead minnow (Cyprinodon variegatus) ☐ Silverside minnow (Menidia beryllina) ☐ Other (write in comments section)			
COMMENT SECTION:										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Jacobo / City of Cadillac	3-10-21	10AM	Mary Wiersma / ERM	03/11/21	10900

* See Instructions for Sample Collection on Back of Sheet

December 2020



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ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:		City of Cadillac		SAMPLER	TL					
ADDRESS:		1121 Plett Rd Cadillac MI		PHONE NUMBER:	231-775-2841					
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (filled in by ERM)				
	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)									
Test	3-12-21	6 AM	Comp	3 gal	pH= 7.04 s.u. NH ₃ = 0.57 mg/L	Temp. (°C) 3 On Ice	D.O. 10.7 mg/L	pH 7.4 s.u.	Cond/Sal 959	µS ppt
	3-12-21	10 AM			pH= s.u. NH ₃ = mg/L	Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt
					pH= s.u. NH ₃ = mg/L	Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt
					pH= s.u. NH ₃ = mg/L	Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt
					pH= s.u. NH ₃ = mg/L	Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal	µS ppt
ANALYSES REQUESTED [check item(s)]	Test Material:		Test Type:	Test Species:	Other (write in comments section)					
	Water/Wastewater Sediment Product				Rainbow Trout (<i>Oncorhynchus mykiss</i>) Sheepshead minnow (<i>Cyprinodon variegatus</i>) Silverside minnow (<i>Menidia beryllina</i>) Fathead minnow (<i>Pimephales promelas</i>) Americamysis bahia Hyalosella azteca Chironomus dilutus					
COMMENT SECTION:										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Lacyko / City of Cadillac	3-12-21	11 AM	D. G. Pl	9/13/21	1200

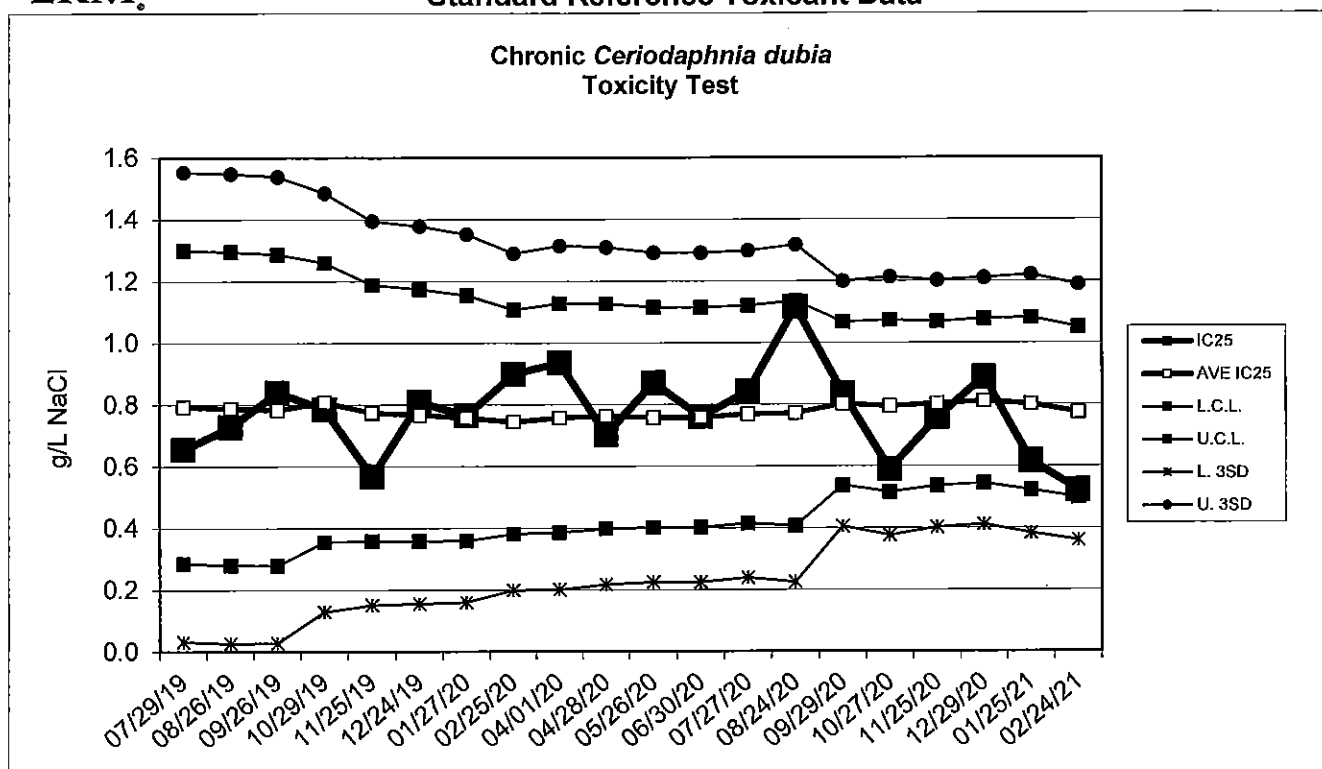
* See Instructions for Sample Collection on Back of Sheet

December 2020



Environmental Resources Management

Standard Reference Toxicant Data



Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT		Survival (%)	CONTROL Reproduction (ave. young)	CV (%)
			Lower	Upper			
07/29/19	0.65	0.8	0.29	1.30	100	33.7	14.6
08/26/19	0.73	0.8	0.28	1.29	100	30.4	23.5
09/26/19	0.84	0.8	0.28	1.29	100	24.8	27.4
10/29/19	0.79	0.8	0.35	1.26	100	37.1	15.9
11/25/19	0.57	0.8	0.36	1.19	100	28.4	30.7
12/24/19	0.81	0.8	0.36	1.17	100	38.5	9.6
01/27/20	0.76	0.8	0.36	1.15	100	21.0	20.3
02/25/20	0.90	0.7	0.38	1.11	100	27.5	14.5
04/01/20	0.94	0.8	0.39	1.13	100	38.2	15.5
04/28/20	0.70	0.8	0.40	1.13	100	33.5	13.7
05/26/20	0.87	0.8	0.40	1.11	100	30.6	19.8
06/30/20	0.76	0.8	0.40	1.11	100	29.8	20.8
07/27/20	0.84	0.8	0.41	1.12	100	31.1	13.5
08/24/20	1.12	0.8	0.41	1.14	100	31.0	13.5
09/29/20	0.84	0.8	0.54	1.07	100	27.7	20.3
10/27/20	0.59	0.8	0.52	1.07	100	31.7	13.7
11/25/20	0.76	0.8	0.54	1.07	100	25.6	24.1
12/29/20	0.89	0.8	0.54	1.08	100	26.6	10.5
01/25/21	0.62	0.8	0.52	1.08	100	21.1	30.2
02/24/21	0.52	0.8	0.50	1.05	100	26.6	19.8

Appendix B
MDEGLE Reporting Form



MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than 1 test.

1. NAME OF FACILITY (on NPDES permit) City of Cadillac WWTP		2. NPDES PERMIT # M I 0 0 2 0 2 5 7							
1. RECEIVING WATER (as designated in permit) Clam River		2. OUTFALL 001A		5. RECEIVING WATER CONCENTRATION (if known)					
6. TEST LAB (Name and Address) Environmental Resources Management 3352 128 th Avenue Holland, MI 49424-9263									
7. TEST START DATE 03/09/21		8. TEST END DATE 03/15/21		9. AGE RANGE OF ORGANISMS AT TEST START 16 – 24 Hours Old			10. REPORT DATE 03/25/21		
11. NAME OF PERSON CONDUCTING TEST Lab Contact: Mark Baker		12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Bruce Rabe (616) 738 – 7308							
13. SAMPLE COLLECTION DATES Sample 1: 03/08/21 Sample 2: 03/10/21 Sample 3: 03/12/21		14. DATE RECEIVED Sample 1: 03/09/21 Sample 2: 03/11/21 Sample 3: 03/13/21			15. ARRIVAL TEMP (C) Sample 1: 1 Sample 2: 1 Sample 3: 3				
16. DATE OF FIRST USE Sample 1: 03/09/21 Sample 2: 03/11/21 Sample 3: 03/13/21		17. TOTAL RESIDUAL CHLORINE (mg/l) Sample 1: <0.01 Sample 2: <0.01 Sample 3: <0.01			18. AMMONIA (mg/l as N) Sample 1: 0.06 Sample 2: 0.07 Sample 3: 0.09				
19. WAS SAMPLE DECHLORINATED? Sample 1: ☐ YES ☒ NO IF YES ➡ Sample 2: ☐ YES ☒ NO IF YES ➡ Sample 3: ☐ YES ☒ NO IF YES ➡		20. DESCRIBE DECHLORINATION (if any)							
21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE									
22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time.) N/A									
23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡					24. STATE MESH SIZE OF FILTER				
25. EFFLUENT SAMPLE TYPE (check one type for each sample) Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE							26. IDENTIFY THE DILUENT (O ₁) CONTROL <u>RMHW</u> IDENTIFY THE SECONDARY (O ₂) CONTROL (if used) _____		
27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION									
CONCENTRATION OF EFFLUENT (%)		O ₁	O ₂	13%	25%	50%	76%	100%	
48-HOUR SURVIVAL (%)		100	N/A	100	100	100	100	100	
6-DAY MEAN REPRODUCTION/FEMALE		30.2	N/A	38.5	32.8	36.6	33.6	27.7	
6-DAY MEAN SURVIVAL (%)		100	N/A	100	100	100	100	100	
28. 48-HOUR LC ₅₀ (%) >100			29. TU _a (acute toxic units) 0						
30. 6-DAY CHRONIC VALUE (%) >100		31. NOEC 100%		32. LOEC >100%		33. TU _c (chronic toxic units) 0			

EQP 5819 (02/10)

FINAL REPORT

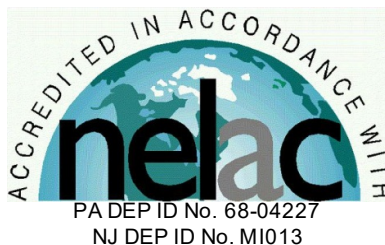
Chronic Toxicity Test
Freshwater Vertebrate,
Pimephales promelas
EPA Test Method 1000.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 9 – 16 March 2021

Laboratory I.D. Number: 030921-6



Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424



Test Overview



Permittee: City of Cadillac WWTP
Location: 1121 Plett Rd.
Cadillac, MI 49601
Contact: Mr. Doug Langworthy
Telephone #: 231.878.4437

NPDES Permit #: MI0020257
Permit Requirements: Reporting Criteria – 1.0 TUa, 1.0 TUc
Test Sample: Outfall 001A
Receiving Water: Clam River

Testing Date: 9 – 16 March 2021

Sample Date(s): 8 March 2021
10 March 2021
12 March 2021

Test/Method: Fathead Minnow,
Pimephales promelas,
Survival and Growth Test
EPA 821-R-02-013
Method 1000.0.

QC Objectives: Test data met all test
acceptability criteria, except
where noted below.

Data Qualifiers: None

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Growth Average Wt./ Organism (mg)
Control	100	0.635
13	100	0.618
25	97.5	0.635
50	97.5	0.636
76	97.5	0.635
100	92.5	0.617

TEST RESULTS

96-Hour LC ₅₀	>100%
NOEC	100%
LOEC	>100%
ChV	>100%
MSDp (Growth)	16.8%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	0

TEST CONCLUSION

In accordance with the NPDES permit requirements for the City of Cadillac WWTP, this toxicity test did not exhibit either acute or chronic toxicity.

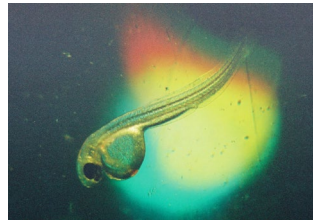
Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0550183.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777



ERM Testing Method

Pimephales promelas – Survival and Growth Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (μm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius ($^{\circ}\text{C}$) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Pimephales promelas used to initiate this test were obtained from in-house cultures and were less than 24-hours old at test initiation. Test organisms were maintained in reconstituted moderately hard water prior to test initiation.

The *Pimephales promelas* test was conducted using 300 to 500-milliliter (mL) disposable polypropylene containers containing 250 mL of control water or test solution. Ten fish were randomly added to each test chamber with four replicate chambers per treatment. Each *Pimephales promelas* test chamber was fed 0.1 mL of a concentrated suspension of less than 24-hour old live brine shrimp nauplii (*Artemia* sp.) two times per day. Test solutions were renewed daily during the exposure by replacing approximately 90 percent of the 24-hour old solution with fresh control water or appropriate test solution. Prior to test solution renewal, uneaten and dead brine shrimp, along with other debris, were removed from the bottom of the test chambers.

Percent survival of exposed *Pimephales promelas* was determined daily by enumeration of live organisms. Mortality was defined as no body movement after gentle prodding. At the termination of the chronic test, larvae in each test chamber were counted, dried, and weighed to the nearest 0.01 milligram (mg) on an analytical balance.

The test was conducted at a temperature of $25 \pm 1^{\circ}\text{C}$ under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Pimephales promelas* survival and growth. An NOEC is defined as the highest effluent concentration which does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration which does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration.

Prior to the determination of any significant differences in *Pimephales promelas* survival and growth, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed, followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or growth) and converted to a chronic toxic unit (TUC) for reporting purposes by $100/\text{ChV}$, unless the LOEC was greater than 100 percent in which case the TUC was reported as 0.

To evaluate acute toxicity, a 96-hour LC₅₀ and corresponding 95 percent confidence interval were also calculated, where possible. The LC₅₀ value estimate was determined by using one of the following statistical methods: graphical, Spearman-Kärber, Trimmed Spearman-Kärber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 96-hour LC₅₀ value was converted to an acute toxic unit (TUa) by 100/LC₅₀.

For 96-hour LC₅₀ values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in the 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.7.6 software program

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC₂₅) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC₂₅ of a particular reference toxicant test does not fall within the expected range of $\pm$ two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

**Environmental
Resources
Management**

***Pimephales promelas* - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation**

Table 1
Page 1 of 1

Permittee/Client: City of Cadillac WWTP

Effluent/Location: Outfall 001A

Lab I.D.#: 030921-6

Beginning Date: 03/09/21

Ending Date: 03/16/21

Time: 1500

Time: 1430

Control/Dilution Water: RMHW

Organism Batch #: 34-21

Organism Age: 224 hrs.

QC Review: VS

QC Review Date: 03/23/21

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	030921-6	031121-3	031321-8	26-21	33-21	--
Temperature	° C	1	1	3	--	--	--
Dissolved Oxygen	mg / L	10.6	11.2	10.7	--	--	--
pH	S.U.	7.1	7.5	7.4	8.2	6.1	--
Conductivity	umhos/cm	1056	942	959	316	317	--
Alkalinity	mg / L CaCO ₃	66	94	120	62	62	--
Hardness	mg / L CaCO ₃	240	220	240	80	64	--
Total Ammonia	mg / L NH ₃	0.06	0.07	0.09	--	--	--
Total Residual Chlorine	mg / L Cl ₂	<0.01	<0.01	<0.01	<0.01	<0.01	--
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	MS	MS	BAR	MB	SIR	--

Test Solution Preparation: Test Solution Prepared For Both Species.

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)	Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200	0	TMB	1	26-21
13%	156	1044	1	TMB	1	26-21
25%	300	900	2	MH	2	26-21
50%	600	600	3	TMB	2	26-21
76%	912	288	4	BAR	3	26-21 33-21
100%	1200	0	5	BAR	3	33-21
			6	MH	3	33-21
			7	--	--	--

**Environmental
Resources
Management**

**Pimephales promelas - Chronic Toxicity Test
Water Quality Data**

Table 2
Page 1 of 1

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 030921-6

Water Quality Data:

Dissolved Oxygen (mg/L)														
Day														
Meter #	3	5	3	5	3	5	3	3	3	3	3	5	3	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.5	7.0	8.4	6.9	8.1	7.1	8.2	7.3	8.5	7.3	8.4	7.1	8.4	7.9
13%	8.5	7.2	8.4	6.2	8.2	6.9	8.2	7.2	8.4	7.4	8.5	7.4	8.5	8.1
25%	8.5	6.9	8.4	6.3	8.1	6.6	8.3	6.8	8.4	7.0	8.5	6.7	8.5	7.6
50%	8.4	7.0	8.4	6.8	8.1	6.6	8.2	6.5	8.4	6.7	8.5	6.3	8.4	6.9
76%	8.4	7.0	8.4	6.4	8.2	6.4	8.2	6.5	8.4	6.4	8.5	6.0	8.4	6.7
100%	8.4	6.9	8.4	5.9	8.3	6.6	8.2	6.3	8.4	6.3	8.5	5.7	8.4	6.3
pH (S.U.)														
Day														
Meter #	9	10	9	9	10	9	10	6	10	10	10	10	9	10
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.1	7.9	8.0	7.9	8.1	7.9	8.0	7.7	8.1	7.8	8.6	8.0	8.0	8.1
13%	--	7.9	--	7.7	--	7.9	--	7.7	--	7.9	--	7.9	--	8.0
25%	--	7.9	--	7.7	--	7.9	--	7.7	--	7.6	--	7.8	--	8.0
50%	--	7.8	--	7.6	--	7.9	--	7.7	--	7.8	--	7.8	--	7.9
76%	--	7.8	--	7.7	--	7.9	--	7.8	--	7.8	--	7.8	--	8.0
100%	8.74	7.8	7.3	7.6	7.7	8.0	7.7	7.8	7.6	7.9	7.4	7.9	7.5	7.9
Conductivity (umhos / cm)														
Day														
Meter #	4	--	4	--	4	--	4	--	4	--	4	--	4	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	311	--	313	--	312	--	315	--	317	--	321	--	317	--
13%	387	--	386	--	387	--	390	--	403	--	409	--	401	--
25%	455	--	455	--	456	--	455	--	481	--	488	--	475	--
50%	595	--	595	--	595	--	596	--	640	--	651	--	640	--
76%	763	--	764	--	764	--	741	--	832	--	820	--	--	--
100%	844	--	876	--	844	--	870	--	984	--	1000	--	--	--
Temperature (°C)														
Day														
Meter #	3	5	3	5	3	5	3	3	3	3	3	5	3	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	25	24	25	24	25	24	24	25	24	25	25	24	25
13%	24	25	24	25	24	25	24	24	25	24	25	25	24	25
25%	25	25	24	25	25	25	25	24	25	24	25	25	24	25
50%	25	25	24	25	25	25	25	24	25	24	25	25	24	25
76%	26	25	24	25	25	25	25	24	25	24	25	25	24	25
100%	26	25	24	25	25	25	26	24	25	24	25	25	24	25
I = Initial Chemistry F = Final Chemistry														

Note: D.O. meter also used for temperature measurement unless otherwise noted.

**Environmental
Resources
Management**

***Pimephales promelas* - Chronic Toxicity Test
Survival Data**

Table 3
Page 1 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 030921-6

Survival Data:

Treatment (% Effluent)	Rep.	# Live Organisms Day								Rep.	# Live Organisms Day								96 Hour Survival Summary		
		0	1	2	3	4	5	6	7		0	1	2	3	4	5	6	7	Total Live		%
		Initial	Final	Initial	Final	Initial	Final	Initial	Final		Initial	Final	Initial	Final	Initial	Final	Initial	Final			
Control	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100
13%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100
25%	A	10	10	10	10	10	9	9	9	B	10	10	10	10	10	10	10	10	40	40	100
50%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	9	9	9	9	40	39	97.5
76%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	39	97.5
100%	A	10	9	9	9	8	8	8	8	B	10	10	10	10	10	10	10	10	40	37	92.5

Treatment (% Effluent)	Rep.	# Live Organisms Day								Rep.	# Live Organisms Day								7 Day Survival Summary		
		0	1	2	3	4	5	6	7		0	1	2	3	4	5	6	7	Total Live		%
		Initial	Final	Initial	Final	Initial	Final	Initial	Final		Initial	Final	Initial	Final	Initial	Final	Initial	Final			
Control	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100
13%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100
25%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	39	97.5
50%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	39	97.5
76%	C	10	10	10	10	10	10	10	10	D	10	10	9	9	9	9	9	9	40	39	97.5
100%	C	10	10	10	10	10	10	10	10	D	10	10	9	9	9	9	9	9	40	37	92.5

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Time:	1500	1430	1500	1430	1330	1300	1400	1430
Initials:	MS	MS	MS	MS	BM	BM	MS	MS
Date:	03/09/21	03/10/21	03/11/21	03/12/21	03/13/21	03/14/21	03/15/21	03/16/21

Feeding:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Batch #:	67-21	68-21	69-21	70-21	71-21	72-21	73-21	--
Initials AM:	--	MS	MS	MS	MS	MS	MS	--
Initials PM:	MS	MS	BM	BM	BM	BM	BM	--

Oven:

Date In	Time In	Initials	Date Out	Time Out	Initials
03/16/21	1430	MS	03/17/21	1500	MS

Comment Section:

Day	Date	Initials	Comments

**Environmental
Resources
Management**

**Pimephales promelas - Chronic Toxicity Test
Growth Data**

Table 3
Page 2 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 030921-6

Pan #	Conc. (% Effluent)	Replicate	Final Weight (mg)	Initial Weight (mg)	Larvae Weight (mg)	# of Initial Organisms	Avg. Wt./ Organism/ Replicate (mg)	Avg. Wt./ Organism/ Treatment (mg)	Avg. Wt./ Organism/ Treatment % CV
Date			3/18/2021	3/15/2021					
Analyst			sb	sb					
C-1	Control	A	20.28	13.60	6.68	10	0.668		
C-2	Control	B	21.61	14.81	6.80	10	0.680		
C-3	Control	C	23.78	18.18	5.60	10	0.560		
C-4	Control	D	21.58	15.26	6.32	10	0.632	0.635	8.5
1-1	13%	A	24.44	18.54	5.90	10	0.590		
1-2	13%	B	24.82	17.80	7.02	10	0.702		
1-3	13%	C	22.10	16.62	5.48	10	0.548		
1-4	13%	D	24.25	17.94	6.31	10	0.631	0.618	10.6
2-1	25%	A	22.89	16.22	6.67	10	0.667		
2-2	25%	B	21.31	15.21	6.10	10	0.610		
2-3	25%	C	22.38	15.40	6.98	10	0.698		
2-4	25%	D	26.41	20.77	5.64	10	0.564	0.635	9.4
3-1	50%	A	23.63	17.68	5.95	10	0.595		
3-2	50%	B	22.85	15.70	7.15	10	0.715		
3-3	50%	C	25.28	19.18	6.10	10	0.610		
3-4	50%	D	21.73	15.49	6.24	10	0.624	0.636	8.5
4-1	76%	A	23.64	18.06	5.58	10	0.558		
4-2	76%	B	25.51	18.38	7.13	10	0.713		
4-3	76%	C	21.34	14.39	6.95	10	0.695		
4-4	76%	D	24.88	19.16	5.72	10	0.572	0.635	12.7
5-1	100%	A	24.59	19.02	5.57	10	0.557		
5-2	100%	B	26.04	20.16	5.88	10	0.588		
5-3	100%	C	23.42	17.11	6.31	10	0.631		
5-4	100%	D	25.91	19.00	6.91	10	0.691	0.617	9.4

Quality Assurance

Final Wt. (mg)

A	Blank	A	18.98	19	-0.02
B	Blank	B	19.75	19.74	0.01

CETIS Analytical Report

Report Date: 17 Mar-21 14:19 (p 1 of 2)
Test Code/ID: 2335968B / 05-9071-4507

Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 07-1590-3317	Endpoint: 7d Survival Rate	CETIS Version: CETISv1.9.7
Analyzed: 17 Mar-21 14:19	Analysis: Nonparametric-Control vs Treatments	Status Level: 1
Edit Date: 17 Mar-21 14:18	MD5 Hash: A1F4E4022EC7A27C4123FC65A7D159EB	Editor ID: 009-972-616-9
Batch ID: 10-7168-3253	Test Type: Growth-Survival (7d)	Analyst: Lab Tech
Start Date: 09 Mar-21 15:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 16 Mar-21 14:30	Species: Pimephales promelas	Brine:
Test Length: 6d 23h	Taxon: Actinopterygii	Source: In-House Culture Age: <24
Sample ID: 19-0101-1845	Code: 714F2385	Project: Wet Compliance Test
Sample Date: 08 Mar-21 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 09 Mar-21 11:00	CAS (PC):	Station: Outfall 001A
Sample Age: 32h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	MSDu	PMSD
Angular (Corrected)	C > T	100	>100	---	1	0.08754	8.75%

Steel Many-One Rank Sum Test

Control	vs	Conc-%	Test Stat	Critical	Ties	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	18	10	1	6	CDF	0.8333	Non-Significant Effect
		25	16	10	1	6	CDF	0.6105	Non-Significant Effect
		50	16	10	1	6	CDF	0.6105	Non-Significant Effect
		76	16	10	1	6	CDF	0.6105	Non-Significant Effect
		100	14	10	1	6	CDF	0.3451	Non-Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	1	0.8	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.0364972	0.0072994	5	1.055	0.4169	Non-Significant Effect
Error	0.124544	0.0069191	18			
Total	0.161041		23			

ANOVA Assumptions Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variance	Bartlett Equality of Variance Test				Indeterminate
Distribution	Shapiro-Wilk W Normality Test	0.85	0.884	0.0022	Non-Normal Distribution

7d Survival Rate Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.00%	0.00%
13		4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.00%	0.00%
25		4	0.9750	0.8954	1.0000	1.0000	0.9000	1.0000	0.0250	5.13%	2.50%
50		4	0.9750	0.8954	1.0000	1.0000	0.9000	1.0000	0.0250	5.13%	2.50%
76		4	0.9750	0.8954	1.0000	1.0000	0.9000	1.0000	0.0250	5.13%	2.50%
100		4	0.9250	0.7727	1.0000	0.9500	0.8000	1.0000	0.0479	10.35%	7.50%

Angular (Corrected) Transformed Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	1.4120	1.4120	1.4120	1.4120	1.4120	1.4120	0.0000	0.00%	0.00%
13		4	1.4120	1.4120	1.4120	1.4120	1.4120	1.4120	0.0000	0.00%	0.00%
25		4	1.3710	1.2420	1.5010	1.4120	1.2490	1.4120	0.0407	5.94%	2.89%
50		4	1.3710	1.2420	1.5010	1.4120	1.2490	1.4120	0.0407	5.94%	2.89%
76		4	1.3710	1.2420	1.5010	1.4120	1.2490	1.4120	0.0407	5.94%	2.89%
100		4	1.2950	1.0610	1.5290	1.3310	1.1070	1.4120	0.0735	11.35%	8.28%

CETIS Analytical Report

Report Date: 17 Mar-21 14:19 (p 2 of 2)
Test Code/ID: 2335968B / 05-9071-4507

Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 07-1590-3317 Endpoint: 7d Survival Rate CETIS Version: CETISv1.9.7
Analyzed: 17 Mar-21 14:19 Analysis: Nonparametric-Control vs Treatments Status Level: 1
Edit Date: 17 Mar-21 14:18 MD5 Hash: A1F4E4022EC7A27C4123FC65A7D159EB Editor ID: 009-972-616-9

7d Survival Rate Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	1.0000	1.0000	1.0000	1.0000
13		1.0000	1.0000	1.0000	1.0000
25		0.9000	1.0000	1.0000	1.0000
50		1.0000	0.9000	1.0000	1.0000
76		1.0000	1.0000	1.0000	0.9000
100		0.8000	1.0000	1.0000	0.9000

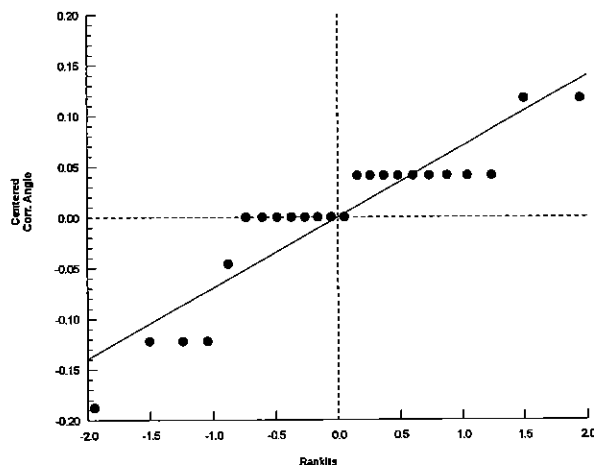
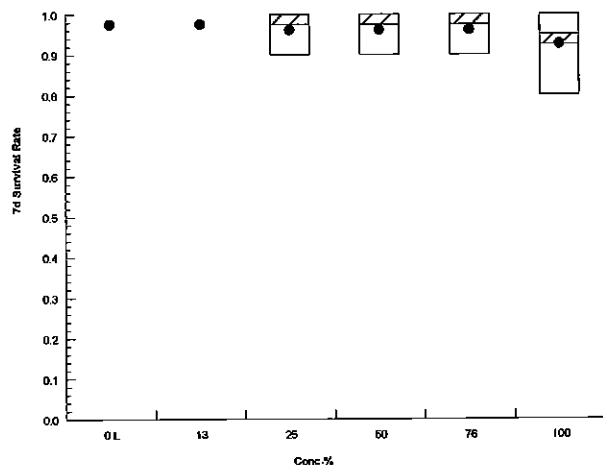
Angular (Corrected) Transformed Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	1.4120	1.4120	1.4120	1.4120
13		1.4120	1.4120	1.4120	1.4120
25		1.2490	1.4120	1.4120	1.4120
50		1.4120	1.2490	1.4120	1.4120
76		1.4120	1.4120	1.4120	1.2490
100		1.1070	1.4120	1.4120	1.2490

7d Survival Rate Binomials

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	10/10	10/10	10/10	10/10
13		10/10	10/10	10/10	10/10
25		9/10	10/10	10/10	10/10
50		10/10	9/10	10/10	10/10
76		10/10	10/10	10/10	9/10
100		8/10	10/10	10/10	9/10

Graphics



CETIS Analytical Report

Report Date: 19 Mar-21 08:32 (p 1 of 2)
Test Code/ID: 2335968B / 05-9071-4507

Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 00-6992-8475	Endpoint: Mean Dry Biomass-mg	CETIS Version: CETISv1.9.7
Analyzed: 19 Mar-21 8:31	Analysis: Parametric-Control vs Treatments	Status Level: 1
Edit Date: 17 Mar-21 14:18	MD5 Hash: BAB6AA5A6EB034718E7367CD7D06F715	Editor ID: 009-972-616-9
Batch ID: 10-7168-3253	Test Type: Growth-Survival (7d)	Analyst: Lab Tech
Start Date: 09 Mar-21 15:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 16 Mar-21 14:30	Species: Pimephales promelas	Brine:
Test Length: 6d 23h	Taxon: Actinopterygii	Source: In-House Culture Age: <24
Sample ID: 19-0101-1845	Code: 714F2385	Project: Wet Compliance Test
Sample Date: 08 Mar-21 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 09 Mar-21 11:00	CAS (PC):	Station: Outfall 001A
Sample Age: 32h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	MSDu	PMSD
Untransformed	C > T	100	>100	---	10	0.1067	16.80%

Dunnett Multiple Comparison Test

Per MSDGLE

Control	vs	Conc-%	Test Stat	Critical	MSD	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	0.3891	2.407	0.107	6	CDF	0.6945	Non-Significant Effect
		25	0.005641	2.407	0.107	6	CDF	0.8317	Non-Significant Effect
		50	-0.02256	2.407	0.107	6	CDF	0.8399	Non-Significant Effect
		76	0.01128	2.407	0.107	6	CDF	0.8300	Non-Significant Effect
		100	0.4117	2.407	0.107	6	CDF	0.6852	Non-Significant Effect

Test Acceptability Criteria

TAC Limits

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	0.635	0.25	>>	Yes	Passes Criteria
PMSD	0.168	0.12	0.3	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.0016994	0.0003399	5	0.08647	0.9935	Non-Significant Effect
Error	0.0707513	0.0039306	18			
Total	0.0724507		23			

ANOVA Assumptions Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variance	Bartlett Equality of Variance Test	0.6768	15.09	0.9842	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.9182	0.884	0.0534	Normal Distribution

Mean Dry Biomass-mg Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	0.635	0.5491	0.7209	0.65	0.56	0.68	0.027	8.50%	0.00%
13		4	0.6177	0.5134	0.7221	0.6105	0.548	0.702	0.0328	10.62%	2.72%
25		4	0.6347	0.5399	0.7296	0.6385	0.564	0.698	0.0298	9.39%	0.04%
50		4	0.636	0.5501	0.7219	0.617	0.595	0.715	0.02699	8.49%	-0.16%
76		4	0.6345	0.5059	0.7631	0.6335	0.558	0.713	0.04039	12.73%	0.08%
100		4	0.6168	0.5244	0.7091	0.6095	0.557	0.691	0.02903	9.41%	2.87%

Mean Dry Biomass-mg Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	0.668	0.68	0.56	0.632
13		0.59	0.702	0.548	0.631
25		0.667	0.61	0.698	0.564
50		0.595	0.715	0.61	0.624
76		0.558	0.713	0.695	0.572
100		0.557	0.588	0.631	0.691

CETIS Analytical Report

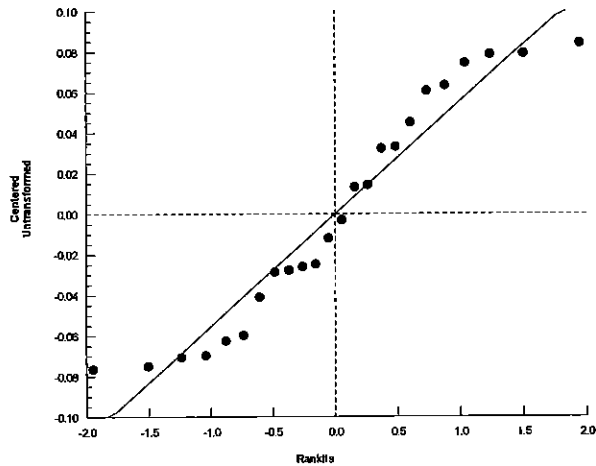
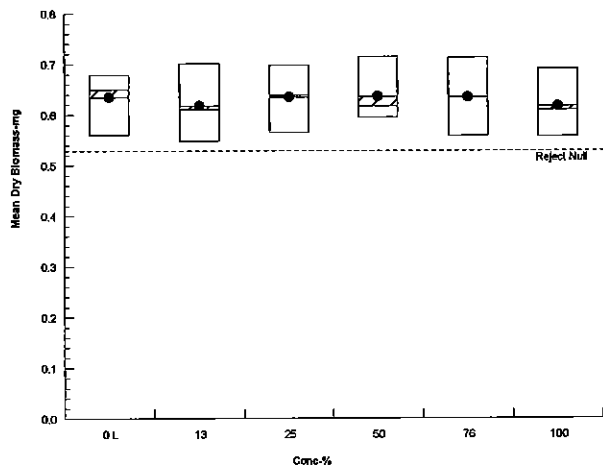
Report Date: 19 Mar-21 08:32 (p 2 of 2)
Test Code/ID: 2335968B / 05-9071-4507

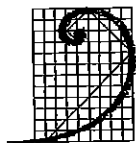
Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 00-6992-8475	Endpoint: Mean Dry Biomass-mg	CETIS Version: CETISv1.9.7
Analyzed: 19 Mar-21 8:31	Analysis: Parametric-Control vs Treatments	Status Level: 1
Edit Date: 17 Mar-21 14:18	MD5 Hash: BAB6AA5A6EB034718E7367CD7D06F715	Editor ID: 009-972-616-9

Graphics





ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

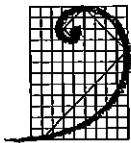
AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac			SAMPLER	TL		
ADDRESS:	1121 Platt Rd Cadillac, MI			PHONE NUMBER:	231-775-2841		
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (Filled in by ERM)
Tert.	3-7-21	7AM	Comp	2 gal.	pH= 6.89 s.u. NH ₃ = .042 mg/L	030921-6	Temp. (°C) 10.6 mg/L D.O. 1056 pH 7.1 s.u. Cond/Sal μS ppt
	3-8-21	7AM			pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal μS ppt
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal μS ppt
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal μS ppt
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal μS ppt
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal μS ppt
					pH= s.u. NH ₃ = mg/L		Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal μS ppt
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product	Test Type: ☒ Acute ☒ Chronic ☐ Other	Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☒ Daphnia pulex ☒ Fathead minnow (Pimephales promelas)	☐ Rainbow Trout (Oncorhynchus mykiss) ☐ Sheepshead minnow (Cyprinodon variegatus) ☐ Silverside minnow (Menidia beryllina)	Other (write in comments section)		
COMMENT SECTION:							

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
Tanya Lageta / City of Cadillac	3-8-21	2pm	Mary Wiershauer / ERM	3/9/21	1100

* See Instructions for Sample Collection on Back of Sheet



ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

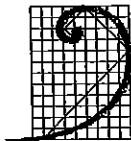
CLIENT NAME: <u>City of Cadillac</u>				SAMPLER <u>TL</u>		PHONE NUMBER: <u>231-775-2841</u>				
ADDRESS: <u>1121 Plett Rd Cadillac, MI</u>										
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
<u>Tert.</u>	<u>3-9-21</u>	<u>7AM</u>	<u>Comp</u>	<u>2gal</u>	<u>pH=7.23 s.u.</u> <u>NH₃=.0553 mg/L</u>	<u>031121-3</u>	Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal µS ppt
	<u>3-10-21</u>	<u>7AM</u>			<u>pH= s.u.</u> <u>NH₃= mg/L</u>		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal µS ppt
					<u>pH= s.u.</u> <u>NH₃= mg/L</u>		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal µS ppt
					<u>pH= s.u.</u> <u>NH₃= mg/L</u>		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal µS ppt
					<u>pH= s.u.</u> <u>NH₃= mg/L</u>		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal µS ppt
					<u>pH= s.u.</u> <u>NH₃= mg/L</u>		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond/Sal µS ppt
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product		Test Type: ☒ Acute ☒ Chronic ☐ Other		Test Species: ☒ <u>Ceriodaphnia dubia</u> ☐ <u>Daphnia magna</u> ☐ <u>Daphnia pulex</u> ☒ <u>Fathead minnow (Pimephales promelas)</u> ☐ <u>Rainbow Trout (Oncorhynchus mykiss)</u> ☐ <u>Sheepshead minnow (Cyprinodon variegatus)</u> ☐ <u>Silverside minnow (Menidia beryllina)</u> ☐ <u>Americanystis bairia</u> ☐ <u>Hyalella azteca</u> ☐ <u>Chironomus dilutus</u> ☐ Other (write in comments section)					
COMMENT SECTION:										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
<u>T. Lagzbo / City of Cadillac</u>	<u>3-10-21</u>	<u>10am</u>	<u>Mary Hirschman / ERM</u>	<u>03/11/21</u>	<u>0900</u>

* See Instructions for Sample Collection on Back of Sheet

December 2020



ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:		City of Cadillac		SAMPLER	TL
ADDRESS:		1121 Platt Rd Cadillac MI		PHONE NUMBER:	231-775-2841
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS
	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)				
Tert	3-12-21	6AM	Comp	3 gal	pH= 7.04 s.u. NH ₃ = 0.578 mg/L Temp. (°C) 3 On Ice D.O. 10.7 mg/L pH 7.4 s.u. Cond/Sal 959 µS ppt
	3-12-21	10AM			pH= s.u. NH ₃ = mg/L Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal µS ppt
030921-6 Pp					pH= s.u. NH ₃ = mg/L Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal µS ppt
					pH= s.u. NH ₃ = mg/L Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal µS ppt
					pH= s.u. NH ₃ = mg/L Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal µS ppt
					pH= s.u. NH ₃ = mg/L Temp. (°C) On Ice D.O. mg/L pH s.u. Cond/Sal µS ppt
ANALYSES REQUESTED [check item(s)]	Test Material:	Test Type:	Test Species:	Other (write in comments section)	
	___ Water/Wastewater ___ Sediment ___ Product	___ Acute ___ Chronic ___ Other	___ Ceriodaphnia dubia ___ Daphnia magna ___ Daphnia pulex ___ Fathead minnow (Pimephales promelas)	___ Rainbow Trout (Oncorhynchus mykiss) ___ Sheepshead minnow (Cyprinodon variegatus) ___ Silverside minnow (Menidia beryllina)	___ Americamysis bahia ___ Hyalella azteca ___ Chironomus dilutus
COMMENT SECTION:					

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Laetke / City of Cadillac	3-12-21	10AM	D. G. Pl	9/13/21	1200

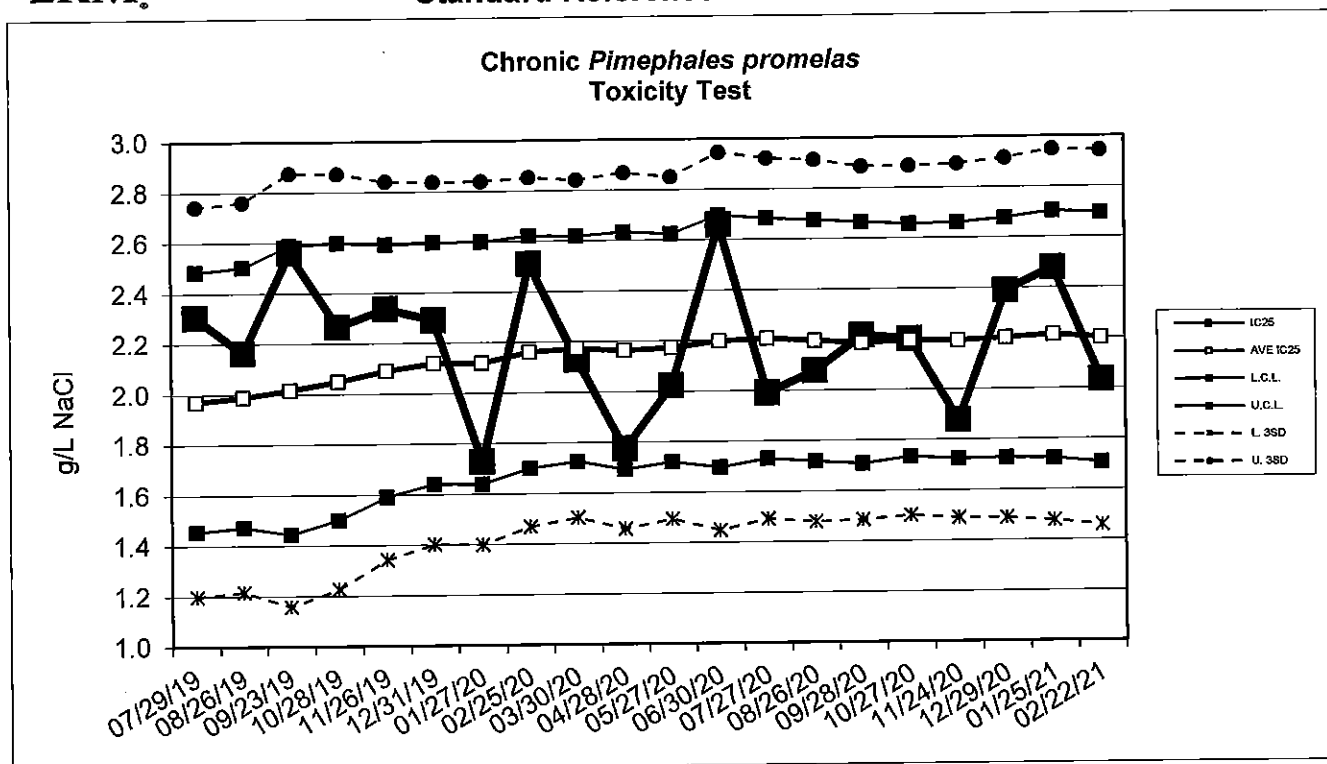
* See Instructions for Sample Collection on Back of Sheet

December 2020



Environmental Resources Management

Standard Reference Toxicant Data



Chronic *Pimephales promelas* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT		Survival (%)	CONTROL Growth (mg)	CV (%)
			Lower	Upper			
07/29/19	2.3	2.0	1.5	2.5	100	0.51	5.6
08/26/19	2.2	2.0	1.5	2.5	100	0.38	15.0
09/23/19	2.6	2.0	1.4	2.6	97.5	0.52	5.8
10/28/19	2.3	2.0	1.5	2.6	100	0.47	5.2
11/26/19	2.3	2.1	1.6	2.6	100	0.51	3.8
12/31/19	2.3	2.1	1.6	2.6	100	0.52	5.1
01/27/20	1.7	2.1	1.6	2.6	97.5	0.71	12.4
02/25/20	2.5	2.2	1.7	2.6	97.5	0.72	7.2
03/30/20	2.1	2.2	1.7	2.6	90	0.52	8.2
04/28/20	1.8	2.2	1.7	2.6	97.5	0.76	11.3
05/27/20	2.0	2.2	1.7	2.6	100	0.54	11.4
06/30/20	2.7	2.2	1.7	2.7	100	0.58	4.1
07/27/20	2.0	2.2	1.7	2.7	97.5	0.38	11.2
08/26/20	2.1	2.2	1.7	2.7	97.5	0.63	11.8
09/28/20	2.2	2.2	1.7	2.7	100	0.62	9.2
10/27/20	2.2	2.2	1.7	2.7	92.5	0.63	6.0
11/24/20	1.9	2.2	1.7	2.7	90	0.42	14.0
12/29/20	2.4	2.2	1.7	2.7	100	0.54	8.4
01/25/21	2.5	2.2	1.7	2.7	97.5	0.83	5.1
02/22/21	2.0	2.2	1.7	2.7	100	0.49	19.2

Appendix B
MDEGLE Reporting Form



MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than one test.

1. NAME OF FACILITY (on NPDES permit) City of Cadillac WWTP				2. NPDES PERMIT # M I 0 0 2 0 2 5 7							
1. RECEIVING WATER (as designated in permit) Clam River				2. OUTFALL 001A				5. RECEIVING WATER CONCENTRATION (if known)			
6. TEST LAB (Name and Address) Environmental Resources Management 3352 128 th Avenue Holland, MI 49424-9263											
7. TEST START DATE 03/09/21			8. TEST END DATE 03/16/21			9. AGE RANGE OF ORGANISMS AT TEST START <24 Hrs Old			10. REPORT DATE 03/25/21		
11. NAME OF PERSON CONDUCTING TEST Lab Contact: Mark Baker						12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Bruce Rabe (616) 738-7308					
13. SAMPLE COLLECTION DATES Sample 1: 03/08/21 Sample 2: 03/10/21 Sample 3: 03/12/21				14. DATE RECEIVED Sample 1: 03/09/21 Sample 2: 03/11/21 Sample 3: 03/13/21				15. ARRIVAL TEMP (C°) Sample 1: 1 Sample 2: 1 Sample 3: 3			
16. DATE OF FIRST USE Sample 1: 03/09/21 Sample 2: 03/11/21 Sample 3: 03/13/21				17. TOTAL RESIDUAL CHLORINE (mg/l) Sample 1: <0.01 Sample 2: <0.01 Sample 3: <0.01				18. AMMONIA (mg/l as N) Sample 1: 0.06 Sample 2: 0.07 Sample 3: 0.09			
19. WAS SAMPLE DECHLORINATED? Sample 1: ☐ YES ☒ NO IF YES ➡ Sample 2: ☐ YES ☒ NO IF YES ➡ Sample 3: ☐ YES ☒ NO IF YES ➡				20. DESCRIBE DECHLORINATION (if any)							
21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE											
22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time. Attach a separate sheet if necessary.) N/A											
23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡						24. STATE MESH SIZE OF FILTER (if filtered)					
25. EFFLUENT SAMPLE TYPE (check one type for each sample) Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) _____ ☐ GRAB SAMPLE Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) _____ ☐ GRAB SAMPLE Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) _____ ☐ GRAB SAMPLE								26. IDENTIFY THE DILUENT (O ₁) CONTROL RMHW IDENTIFY THE SECONDARY (O ₂) CONTROL (if used) _____			
27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND GROWTH											
CONCENTRATION OF EFFLUENT (%)		O ₁ (dilution H ₂ O)	O ₂	13%	25%	50%	76%	100%			
96-HOUR SURVIVAL (%)		100	N/A	100	100	97.5	97.5	92.5			
7-DAY MEAN BIOMASS (mg/initial fish)		0.635	N/A	0.618	0.635	0.636	0.635	0.617			
7-DAY MEAN SURVIVAL (%)		100	N/A	100	97.5	97.5	97.5	92.5			
28. 96-HOUR LC ₅₀ (%) >100			29. TU _a (acute toxic units) 0								
30. 7-DAY CHRONIC VALUE (%) >100		31. NOEC 100%		32. LOEC >100%		33. TU _c (chronic toxic units) 0					

EQP 5820 (02/10)